Higher National School of Advanced Technologies (ENSTA)

SS/IA: First Semester - 2026

**Module: WAMD**

**TD -TP 2: Create a React Application with Redux for State Management**

## 1. Lab Objectives

At the end of this lab, students should be able to:

- Create and run a **React.js application**.
- Understand the basic structure of a React application.
- Build reusable **React components**.
- Use **JSX** to describe user interfaces.
- Manage component data using **props**.
- Manage local state using **React state**.
- Understand the role of **global state management**.
- Configure and use **Redux** in a React application.
- Create a Redux **store**.
- Define **actions** and **reducers**.
- Access and update global state from React components.
- Use **React-Redux** to connect components to the Redux store.
- Organize a React project into appropriate files and components.
- Test and debug a React application using browser Developer Tools.

## 2. Tools and Technologies

| Tool / Technology | Purpose |
|---|---|
| **Visual Studio Code** | Development environment |
| **React.js** | Building the user interface |
| **JavaScript (ES6+)** | Application logic |
| **Redux** | Global state management |
| **React-Redux** | Connecting React components with Redux |
| **Node.js / npm** | Installing dependencies and running the application |
| **Google Chrome / Firefox** | Testing and debugging |
| **Browser DevTools** | Inspecting and debugging the application |
| **Git / GitHub** | Version control and project submission |

## 3. Lab Statement

**Problem**

You are asked to develop a **small e-commerce web application using React.js**. The application represents an online store where users can browse products and manage a simple shopping cart. The main objective is to move from a traditional JavaScript application to a **component-based React application** and understand why **global state management** is useful when several components need to access or modify the same data.

The application must demonstrate the use of:

- React components
- JSX

- Props
- State
- Events
- Redux Store
- Actions
- Reducers
- React-Redux

The application should provide a **clear, responsive and user-friendly interface**. **Important:** This lab focuses on the React frontend and Redux. A backend, database, authentication system and payment system are not required. They will be introduced in later labs.



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## 4. Functional Requirements

The application must:

**React**

- Be developed using **React.js**.
- Use JSX.
- Contain multiple reusable components.
- Use props.
- Use local state where appropriate.
- Handle user events.

**E-Commerce**

- Display a product catalog.
- Display product information.
- Allow users to interact with products.
- Provide a shopping cart.
- Allow products to be added to the cart.
- Allow products to be removed from the cart.
- Dynamically update the interface when the cart changes.

**Redux**

- Use a Redux store.
- Define actions.
- Define reducer logic.
- Use React-Redux.
- Read global state from React components.
- Dispatch actions from React components.
- Display the updated global state.

**Interface**

- Provide a clear and user-friendly interface.

## 5. Optional Features

Students who complete the required functionality may implement additional features. Examples include:

- Product search
- Category filtering
- Product details
- Quantity management
- Cart total
- Favorites

- Product sorting
- Dark/light mode
- Simple navigation
- Local storage
- Discount display

These features are **optional**.
The priority is to correctly implement **React, components, state and Redux**.

**6. Creative Requirement**

The **application theme and visual identity are intentionally left open to creativity**. Students are free to choose:

- Store name
- Logo
- Product category



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- Products
- Colors
- Typography
- Layout
- Images
- Navigation style
- Product-card design
- Cart design
- Additional features

**For example:**
DzTech Store → Electronics
DzFashion → Clothing
BookDZ → Books
SportZone → Sports
ArtisanDZ → Handmade products
Students may propose another appropriate e-commerce concept.
**Creativity is encouraged, but all required technical functionalities must be respected.**

**7. Deliverables**

Students should submit their work through the **course Classroom**.
The submission must contain:

1. **GitHub repository link** containing the complete React project.
2. **Screenshots of the final application**.
3. A README.md containing:
    - Student name
    - Application/store name
    - Application description
    - Technologies used
    - Main React components
    - Redux state managed by the application
    - Implemented features
    - Instructions for running the application

**Optional:** Students may deploy their React application online and provide the live URL. Short demonstration video.

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